Predicting Outcomes in a Football Match Using Statistical Techniques

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Outline

1. Introduction & Motivation
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Why Traditional Metrics Are Incomplete

- Limited to **aggregate count metrics** (number of passes, tackles, possession percentage)
- Context is ignored
- Heavily biased toward terminal events:
  - xG values only *shots*
  - Passes, clearances, duels = 99% of gameplay — yet **invisible** to standard models
- Uses of **rigid, predefined pitch zones** fail to capture fluid team playstyle.

Our Contribution

Novel Framework

Bridge the gap between aggregate metrics and abstract tactical concepts by quantifying *all* football events.

- Discretize pitch into **400 zones** (20×20 grid)
- Apply **unsupervised clustering** using role-based features to identify *functional tactical zones*
- Applied to **2017/18 Manchester City "Centurions"** season (>28,000 passes)
- Validated against contrasting tactical philosophies: **Burnley**

Literature Review

- **Rudd (2011)** — Markov chain for expected goal value
- **Green (2012)** — Introduced **xG** via logistic regression
- **Anzer et al. / Scholtes et al.** — XGBoost & Bayesian xG model
- **Rathke (2017)** — Tactical pitch zones improve xG context
- **Singh (2018)** — **xT framework**: 16×12 grid
- **Gyarmati & Stanojevic** — Markov-chain; 1,000 clusters

Expected Threat (xT) — Singh (2018)

Pitch divided into a $16 \times 12 = 192$ cell grid.
Assumption: Player either **shoot** or **move** the ball. For each cell (x, y):

$m_{x,y}$: probability of *moving* the ball

$s_{x,y}$: probability of *shooting*

$g_{x,y}$: probability a shot becomes a goal

$T_{(x,y)\to(u,v)}$: transition probability

Recursive Value Iteration

- $V_{x,y}$ is expected payoff of cell (x, y)
- Given player shoots, expected payoff is $g_{x,y}$
- Given player moves, expected payoff is $\sum_{(u,v)} T_{(x,y)\to(u,v)} V_{u,v}$
- Total payoff is :

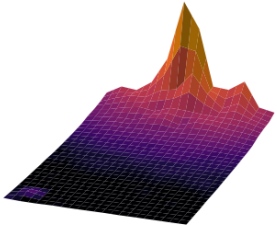
$$V_{x,y} = s_{x,y} \cdot g_{x,y} \; + \; m_{x,y} \cdot \sum_{(u,v)} T_{(x,y)\to(u,v)} V_{u,v}$$

- Iterated to convergence from $V_{x,y} = 0$.

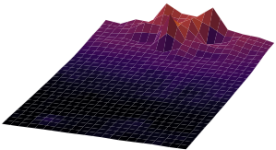
Relation With MDP

This is a **simplified Markov Decision Process** with following specifications:

- **Action space:** 2 actions – shoot and move
- **State space:** 192 pitch zones and 1 absorping state
- **Reward:** $g_{x,y}$ is reward when shot is attempted otherwise 0
- **Objective function:** Cumulative sum of reward for infinite horizon



ManCity xT map (17/18 season)



Swansea xT map (17/18 season)

Source: Karun Singh (2018), Expected Threat Model

Limitations of xT

1. **Uniform threat surface** — ignores team-specific tactics & skill variation
2. **Reward misspecification** — binary $\{0, 1\}$ ignores shot quality
3. **High variance** — frequentist empirical ratios unreliable under data sparsity
4. **No theoretical justification** for infinite horizon ($\gamma = 1$) in finite matches
5. **Pooled training** — single universal transition matrix ignores team heterogeneity

Our objective

Reformulate $V_{x,y}$ so threat surfaces are **statistically significant**, **computationally stable**, and **tactically representative**.

Dataset & Spatial Discretization

- **Dataset:** EPL 2017/18 event data (Pappalardo et al., 2019)
- Wyscout pitch ($x, y \in [0, 100]$) $\rightarrow$ regular **20×20 grid** (400 blocks)
- 10 role based features are prepared from raw event data
- Normalized feature vectors to address volume bias

Role-Based Features (10)

Feature	Role
Fortress	Defensive actions
Launch	Long fwd pass (def half)
Entry	Short fwd pass (att half)
Pivot	Lateral outward pass
Support	Lateral inward pass
Switch	Lateral long pass
Link	Lateral short pass
Safety	Backward pass
Isolation	Att duel / Cross
Finish	Shot

Smoothing & Hellinger distance matrix

1. **Smoothing:** A 5×5 convolution kernel is applied with weights 1, $\frac{1}{2}$, and $\frac{1}{4}$ assigned to the center, first-order, and second-order neighbours, respectively, to mitigate boundary effects and incorporate local spatial context.
2. **Probability vector:** Let $\mathbf{x}_i^{1\times 10}$ denote the feature count vector for the i-th block. Each vector is normalised to a probability distribution $\mathbf{p}_i^{1\times 10}$ by dividing by block's total feature count.
3. **Hellinger distance:** The distance matrix $D_{\text{Hellinger}}$ is constructed such that its (i, j)-th entry represents the Hellinger distance between probability vectors p_i and p_j, defined as:

$$D_H(p_i, p_j) = \frac{1}{\sqrt{2}} \|\sqrt{p_i} - \sqrt{p_j}\|_2$$

Flow distance matrix & Clustering

1. **Direction vector:** For each block **i**, coordinate differences between pass origin and end location are computed and averaged to obtain the mean direction vector $\bar{v}_i$.
2. **Smoothing:** Vectors $\bar{v}_i$ are spatially smoothed via a 3×3 weighted kernel to reduce noise.
3. **Flow distance:** Cosine similarity S_{ij} between $\bar{v}_i$ and $\bar{v}_j$ is converted to a bounded metric:

$$D_{\text{Flow}}(i,j) = \begin{cases} \frac{1-S_{ij}}{2} & \text{if } \bar{v}_i, \bar{v}_j \neq \bar{0} \\ 0.5 & \text{otherwise} \end{cases}$$

4. **Final Distance Matrix:**

$$D_{\text{hybrid}} = \alpha\, D_{\text{Hellinger}} + (1-\alpha)\, D_{\text{Flow}}$$

5. **Agglomerative Hierarchical Clustering** with Complete Linkage with distance threshold $\tau = 0.2$

Validation Framework

Three-Stage Validation

1. **Stability (Split-Half ARI):** Season split into two 19-match halves; Adjusted Rand Index measures cluster agreement
2. **Silhouette Score + Permutation Test:** 1,000 label shuffles; confirms clusters are non-random ($p < 0.01$)
3. **Network Analysis:** Clusters modelled as directed graph $G = (V, E)$
   - Dijkstra shortest paths $\rightarrow$ validates tactical delivery routes
   - Betweenness Centrality $\rightarrow$ identifies key possession junctions
4. **Heatmap Reconstruction:** Cluster ownership mapped back to 20×20 grid; compared to ground-truth KDE

Clustering Result

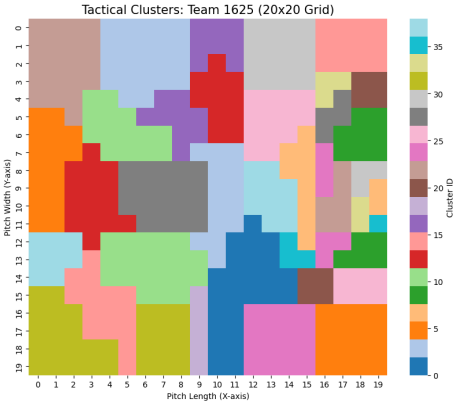


Figure: Clustering results for Manchester City
Reduction from 400 blocks to 39 clusters

Parameter Validation

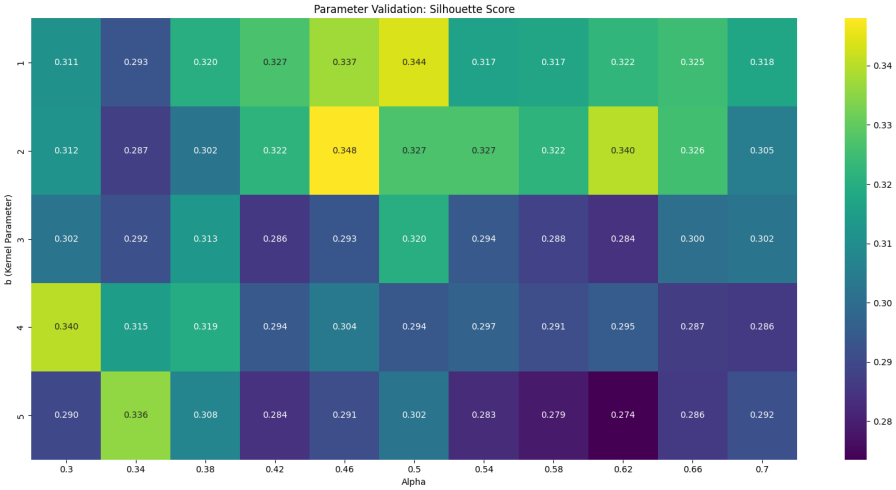
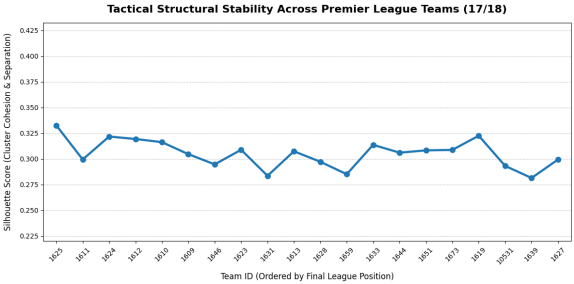


Figure: Best Parameter set $\alpha = 0.46$ and $b = 2$ for Man City

ARI and Silhouette Score



Silhouette Score across all 20 teams

Table: Cluster stability (ARI) across 20 teams

Team	ARI	Team	ARI
Manchester City	0.47	Crystal Palace	0.45
Manchester United	0.48	Bournemouth	0.51
Tottenham	0.48	West Ham	0.43
Liverpool	0.48	Watford	0.46
Chelsea	0.51	Brighton	0.47
Arsenal	0.51	Huddersfield	0.51
Burnley	0.42	Southampton	0.45
Everton	0.47	Swansea City	0.44
Leicester City	0.43	Stoke City	0.44
Newcastle United	0.48	West Bromwich	0.41

Permutation Test

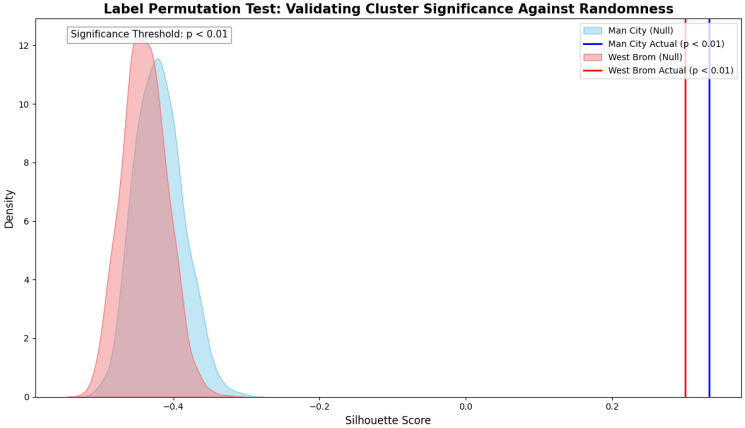
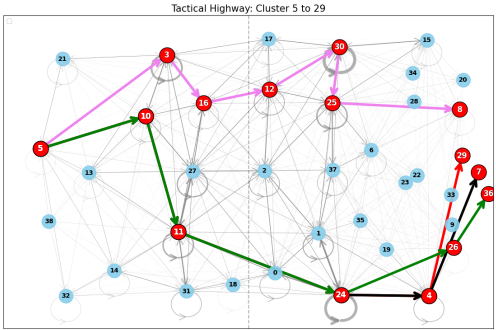
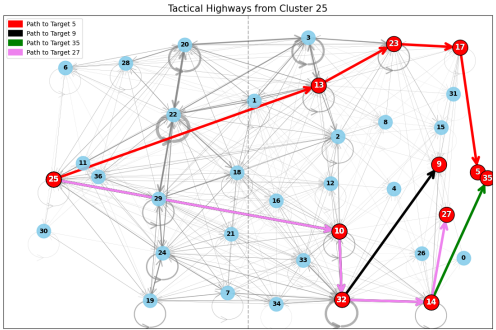


Figure: Test rejected null hypothesis with p-value 0.0009

Transition Flow: Man City vs. Burnley





Manchester City (Positional)

- Gradual build-up through half-spaces
- Ederson → Kompany/Otamendi → Walker/ De Bruyne → Sterling/Silva → Agüero/Jesus

Burnley (Direct)

- Long-ball edges bypass midfield entirely
- GK Pope → offensive aerial zones

Betweenness Centrality — Top 5 Clusters

Man City (Positional)		**Burnley (Direct)**	
Cluster	Centrality	Cluster	Centrality
24 (De Bruyne, Walker)	0.336	22 (Mee)	0.317
11 (Kompany)	0.296	32 (Lowton)	0.261
4 (Kyle Walker)	0.238	24 (Long, Tarkowski)	0.209
25 (D. Silva, De Bruyne)	0.194	3 (Ward)	0.191
1 (Fernandinho)	0.186	23 (Ward)	0.160

Man City: high centrality in *half-space & central pivot* $\Rightarrow$ midfield-driven possession.
Burnley: high centrality in *deep defensive & wide aerial* $\Rightarrow$ defenders as "playmakers".

Heatmap Reconstruction

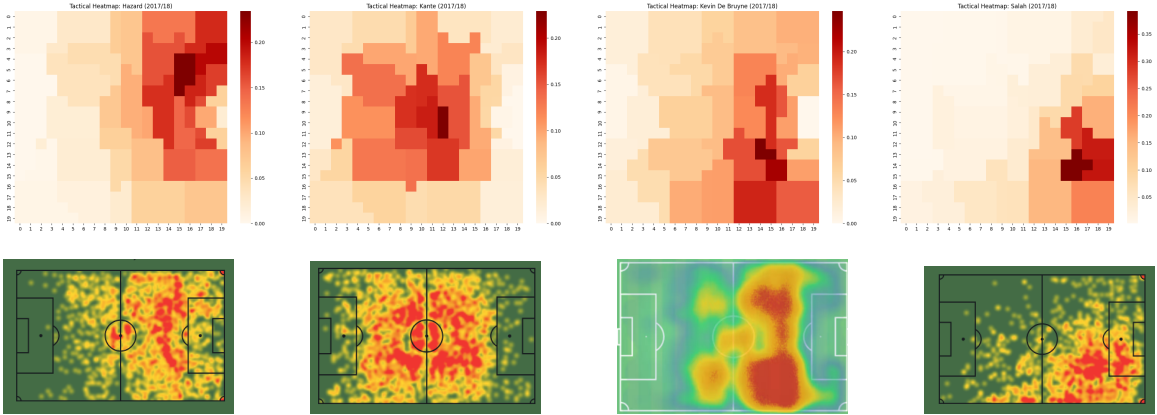


Figure: Comparison of tactical heatmaps vs real heatmaps (2017/18).
From left to right: Hazard, Kante, De Bruyne, Salah.

Summary & Future Work

What We Achieved

- Team-adaptive tactical zones via role-based hierarchical clustering
- Statistically validated clusters ($p < 0.01$, high ARI)
- Network analysis reveals team-specific possession highways
- Heatmap fidelity confirms player-role discrimination

Future Directions

- Incorporate Bayesian value estimation for different events.
- Theoretically justify discount factor $\gamma < 1$ for finite match horizon
- Extend valuation to *all* event types beyond passes

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